

100-degree Non-Contact Retinal Imaging in Mice Using a Multimodal Vis-OCT and SLO System

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Abstract—This study reports a non-contact, multimodal visible-light OCT (vis-OCT) and scanning laser ophthalmoscopy (SLO) system for *in vivo* mouse retinal imaging that achieves a single-shot 100° (visual angle) field of view (FOV). By performing repeated B-scans at the same location, the system enables OCT angiography (OCTA). With a long working distance (6.4 mm), the system also allows ready realignment without disrupting native ocular optics or physiology. We validated system performance in transgenic mice expressing fluorescently labeled microglia, including both healthy mice and oxygen-induced retinopathy (OIR) models. Images from the device clearly resolve individual nerve fiber bundles, inter-plexus capillaries, and individual microglia. Hyaloid vessel shadows and tortuous vessels in OIR mice are also discernible. Together, these capabilities provide a platform for the comprehensive assessment of vascular dysfunction and microglia behavior, offering a promising tool for investigating retinal diseases that affect the peripheral regions.

Index Terms—Visible-light optical coherence tomography, scanning laser ophthalmoscopy, optical coherence tomography angiography, wide-field retinal imaging, retinal microglia.

I. INTRODUCTION

MOUSE models are critical for studying the pathogenesis of retinal and optic nerve disorders and evaluating the efficacy of therapies. Transgenic lines in particular enable in-depth investigation into physiological processes such as microglial activation, an event that occurs before retinal vascular pathology develops, ganglion cell death occurs, or nerve fibers are lost [1].

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To gain deeper insight into the cellular structure and molecular processes of the living retina, non-invasive *in vivo* imaging modalities are highly desirable. Among them, optical coherence tomography (OCT) is widely used to monitor retinal changes [2], [3]. OCT enables three-dimensional, high-resolution structural features, while OCT angiography (OCTA) provides label-free imaging of blood flow and capillary-level vasculature. In addition, compared to the conventional near-infrared OCT (NIR-OCT), visible-light OCT (vis-OCT) offers higher axial resolution with a comparable bandwidth and enables retinal oximetry [4]. Another commonly used imaging modality is scanning laser ophthalmoscopy (SLO) [5], [6]. Beyond two-dimensional structural and vascular imaging, SLO also supports exogenous fluorescence imaging at single-cell resolution, facilitating the study of dynamic behaviors of fluorescently labeled cells [7], [8].

In our previous work, we developed a multimodal vis-OCT and SLO system with a 34° FOV (visual angle) that visualized retinal microstructure, microvasculature, and individual microglia with discernible branches [9]. However, this limited FOV restricts the assessment of the peripheral retina, which is vital for evaluating retinal vascular diseases such as retinopathy of prematurity [10]. To date, the largest FOV reported for OCT/SLO imaging in the mouse retina is 100° (visual angle) [11]. That system integrated NIR-OCT and SLO, enabling OCT, OCTA, SLO reflectance, and SLO fluorescein angiography. Nevertheless, this NIR-OCT provided an axial resolution of only 3.9 μm (in tissue) and a lateral resolution of 10 μm , limiting its ability to resolve fine capillary structures and subtle retinal layer boundaries. vis-OCT achieves a higher resolution, but most vis-OCT systems for mice have a limited FOV ($\sim 50^\circ$, visual angle) [2], [6].

In this Letter, we propose a wide-field, non-contact multimodal vis-OCT and SLO system for mouse retinal imaging. The system achieves a 100° FOV with a theoretical axial and lateral resolutions of 1.6 μm and 5.0 μm in tissue, respectively. In addition, it provides a 6.4 mm working distance, facilitating alignment of the mouse eye, while preserving native ocular optics and physiology. A preliminary version of this work was previously presented in a conference letter [12]. The present Letter expands that work by providing additional details on the system design, evaluation of system performance, and extending its application to the oxygen-induced retinopathy (OIR) study.

II. METHODS

A schematic of our system is presented in Fig. 1. A supercontinuum laser (SuperK COMPACT, NKT Photonics

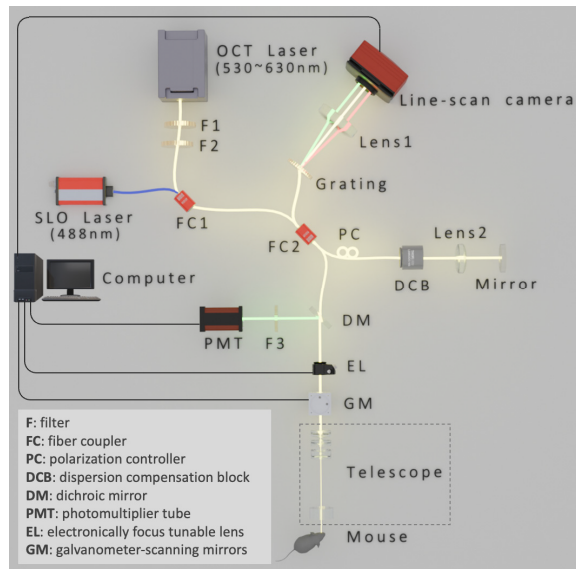


Fig. 1. Multimodal vis-OCT and SLO system schematic diagram. OCT and SLO light sources are first aligned using a fiber coupler (FC1) and then directed into the same sample arm via another fiber coupler (FC2). The back-scattered OCT signal and fluorescence emission signal are separated by a customized dichroic mirror (DM) and subsequently detected by a camera and a photomultiplier tube (PMT), respectively.

Inc), a long-pass filter, and a short-pass filter provided a central wavelength of 580 nm with a bandwidth of ~ 100 nm illumination for the OCT system. A 488 nm laser (0488L-15A, Integrated Optics) was used for the SLO system. Both light sources were connected to a 90/10 single-mode optical fiber coupler for co-alignment, and the output light was divided into a sample and reference arm by another 90/10 fiber coupler. In the reference arm, a fiber polarization controller and a dispersion compensation block were used to alter the polarization mismatch and dispersion mismatch between the two arms. In the sample arm, an electronically focus tunable lens was located before a 2D galvanometer-scanner to change the focal plane in the sample. The telescope consisted of three achromatic lenses ($2 \times$ AC254-075-A, ACN254-040-A, Thorlabs Inc) and a slit lamp lens (90D, Volk Optical Inc). These lenses contributed to a magnification factor of 8.6, amplifying the optical angle from 12° (provided by the scanners) to 100° before entering the pupil, and the beam size was reduced to 0.23 mm on the pupil plane. The OCT back-scattered signal and SLO fluorescence emission signal were separated by a customized dichroic mirror. Then, the OCT signal was directed to a commercial spectrometer (Cobra Vis, Wasatch Photonics) with a line rate of 50 kHz, and OCT volumes were sampled at $2048 \times 1200 \times 1200$ points, thus it took 28.8 s to get an OCT volume. The theoretical axial and lateral OCT resolutions were $1.6 \mu\text{m}$ (in tissue) and $5.0 \mu\text{m}$, respectively. Meanwhile, after passing through a bandpass filter, the SLO signal was collected by a collimator connected to a multimode fiber, then directed to a photomultiplier tube (H7827-002, Hamamatsu Photonics). Theoretically, SLO can achieve a theoretical lateral resolution of $4.2 \mu\text{m}$. Since OCT and SLO share the same scanning optics, they are inherently coarsely aligned; however,

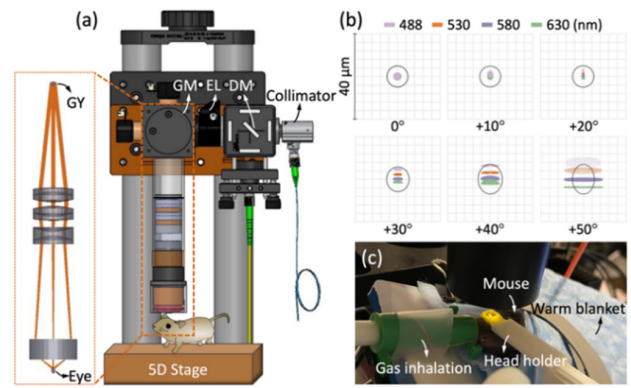


Fig. 2. (a) Mechanical and optical layout of the sample arm and SLO detection path. DM: dichroic mirror; EL: electronically focus tunable lens; GM: galvanometer-scanning mirrors; GY: galvanometer for slow (y) axis. The telescope consists of four lenses, as shown in the inset. (b) Spot diagrams of the optical design, with OCT light (530, 580, and 630 nm) and SLO light (488 nm) for different FOVs. Black circles represent the Airy disk with a $5.0 \mu\text{m}$ radius. (c) The experimental setup illustrates how the mouse eye is scanned.

no explicit pixel-wise co-registration was performed in this study.

To ensure precise alignment of optical components and accurate component locations, we used SolidWorks for the mechanical design (Fig. 2(a)). All components were assembled from off-the-shelf parts, with spacers and holders fabricated using a 3D printer. The sample arm was simulated in OpticStudio. Spot diagrams of OCT light and SLO light at different scanning angles are shown in Fig. 2(b). While optical aberrations are effectively controlled within 80° , they become noticeable at the edge, which may reduce lateral resolution.

OCT data were generated by standard post-processing steps in MATLAB, and *en face* OCT was obtained using mean intensity projection. To get OCTA data, three B-scans were repeated at the same location, and *en face* OCTA was obtained using maximum intensity projection. The OCT system employs the same laser source and spectrometer as our previously reported 34° FOV system [9], where sensitivity, axial resolution, imaging depth, and sensitivity roll-off were validated. Therefore, the on-axis performance of this OCT system is expected to be comparable. Due to the limited imaging range, the OCT B-scan is flipped at the edge, and a dispersion encoded full range algorithm was used to correct these mirroring artifacts [13]. To evaluate the impact of transverse chromatic aberration across the full scanning area, we performed a sub-band analysis of the OCT signal [14].

Five transgenic mice (B6.129P2-Cx3cr1tm1Litt/J/J) with Enhanced Green Fluorescent Protein (EGFP)-labeled retinal microglia were used in this study. Three of them were healthy adult mice, one was an OIR mouse, and the other was an age-matched control. OIR was induced using a 50/10 oxygen exposure paradigm [3]. Before imaging, mice were anesthetized with isoflurane, and the eyes were dilated. During imaging, mice were immobilized on a customized rotation stage. A warm blanket circulated with 37°C water was used to maintain body temperature, and a homemade holder was placed on the head to reduce motion artifacts (Fig. 2(c)). All animal procedures complied with the Association for Research

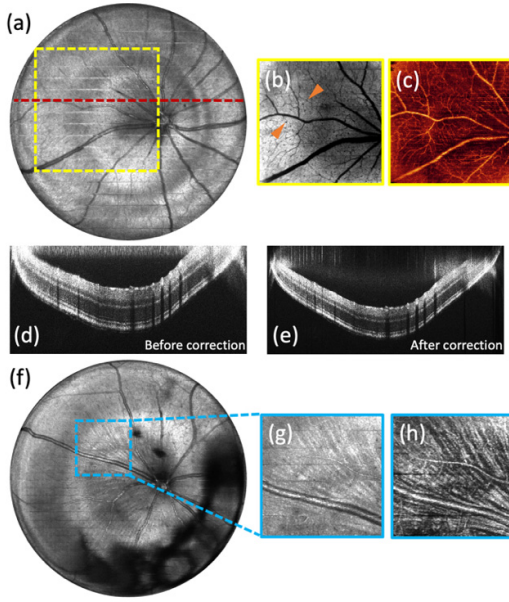


Fig. 3. (a) *En face* OCT of the retina from a mouse, demonstrating the optic disc and major blood vessels across a 100° FOV. (b) *En face* OCT of the outer retina in (a), showing inter-plexus capillaries as black dots (orange arrows). (c) *En face* OCTA of the inner retina in (a), revealing the details of retinal vascular anatomy. (d) OCT B-scan located at the position shown in (a), with mirroring artifacts at the edge. (e) Dispersion encoded full range OCT B-scan at the same position. (f) *En face* OCT of the retina from another mouse. (g) The enlarged image of the marked area in (f). (h) *En face* OCT of the nerve fiber layer within the marked area in (f), with clear differentiation of individual nerve fiber bundles.

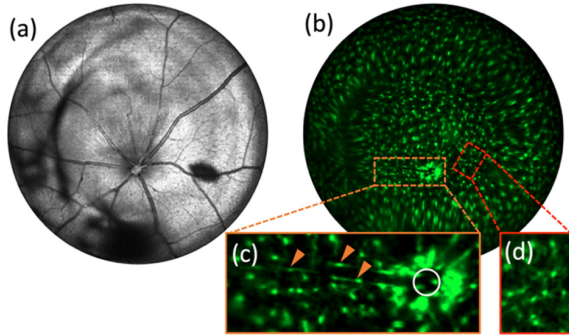


Fig. 4. (a) Sequential OCT (a) and SLO (b) images from a mouse, showing the distribution of retinal vessels and fluorescently labeled microglia. (c) An enlarged image of the orange-marked area in (b), with a white circle indicating the optic disc and orange arrows showing microglia branches along blood vessel walls. (d) An enlarged image of the red-marked area in (b) indicates homeostatic microglia with multiple ramified processes.

in Vision and Ophthalmology Statement and were approved by the Institutional Animal Care and Use Committee at Oregon Health & Science University.

III. EXPERIMENTS AND RESULTS

Retina images of healthy adult mice (~10-week-old, $N = 3$) with a 100° FOV are demonstrated in Figs. 3 and 4. The *en face* OCT of the whole retina depicts the network of major blood vessels (Figs. 3(a) and 3(f)). The outer-retina *en face* OCT (Fig. 3(b)) and inner-retina *en face* OCTA (Fig. 3(c)) reveal inter-plexus capillaries (orange arrows) and the organization of retinal circulation, respectively. In addition, the mirroring artifacts in the OCT B-scan are effectively

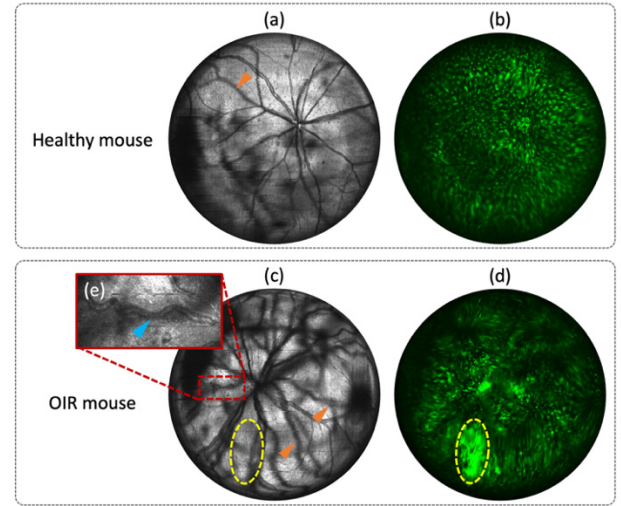


Fig. 5. OCT (a) and SLO (b) images of a healthy young mouse at postnatal day 22, and corresponding images (c, d) from an age-matched OIR mouse. Hyaloid vessel shadows are visible in both OCT images (orange arrows), with substantially more shadows observed in the OIR mouse. In the SLO images, fluorescence from microglia is partially obscured by hyaloid vessels in both animals. The healthy mouse shows a relatively uniform microglial distribution (b), whereas the OIR mouse exhibits a microglial cluster (yellow dashed circle in d) in a region where few hyaloid vessel shadows are present (yellow dashed circle in c). (e) Enlarged view of the highlighted area in (c) reveals a tortuous vessel (blue arrow).

corrected (Fig. 3(e)). The transverse chromatic aberration analysis shows that features' intensity profiles remained spatially aligned across all OCT sub-bands (Supplementary Fig. S1). Moreover, the high resolution of our system is further illustrated in Figs. 3(f)-(h). While fine striations corresponding to the retinal nerve fiber layer are evident in the *en face* OCT of the whole retina (Figs. 3(f) and 3(g)), the individual nerve fiber bundles with clear separation and radial orientation are resolved in the projection of the nerve fiber layer (Fig. 3(h)). The horizontal line artifacts observed in the *en face* OCT are primarily attributable to respiration, which becomes more pronounced with the extended acquisition time.

The system successively captured OCT and SLO from another mouse eye (Fig. 4). The *en face* OCT depicts the retinal structure (Fig. 4(a)), and the SLO image visualizes individual EGFP-labeled microglia within the same FOV (Fig. 4(b)). Furthermore, microglia around the optic disc are clearly resolved, with some branches extending along blood vessel walls. Thanks to the high resolution, homeostatic microglia with multiple ramified processes are also resolved (Fig. 4(d)).

Using this system, we also imaged an OIR mouse and an age-matched healthy control at postnatal day 22. *En face* OCT reveals hyaloid vessel shadows in both eyes (Figs. 5(a) and 5(c)), with a noticeably greater number in the OIR eye. Tortuous vessels within the OIR eye are also detected in OCT (Fig. 5(e)). SLO shows fluorescence signals from microglia, some of which are obscured by the overlying hyaloid vessels (Figs. 5(b) and 5(d)). While the healthy mouse exhibits a relatively uniform microglial distribution (Fig. 5(b)), the OIR mouse shows a microglial cluster (Fig. 5(d)) in a region with comparatively few hyaloid vessels (Fig. 5(c)).

IV. DISCUSSION AND CONCLUSION

We demonstrated high-quality vis-OCT/OCTA and SLO imaging across a 100° FOV in mouse retinas. The system achieves high axial and lateral resolution, and the multimodal configuration allows complementary visualization of retinal structure, vasculature, and fluorescently labeled microglia. Using this system, we imaged five mice. In the OIR mouse model, pathological features such as tortuous vessels are clearly visualized (Figs. 5(c) and 5(e)). Compared to the age-matched healthy control (Fig. 5(b)), the OIR mouse exhibits a relatively uneven microglial distribution (Fig. 5(d)), including a distinct cluster in the periphery. Since the corresponding OCT contained few hyaloid vessels (Fig. 5(c)), this fluorescence aggregation is unlikely to reflect scattering or obscuration by the overlying hyaloid vessels. Instead, the cluster may suggest microglial activation and migration in response to the OIR pathology. However, *ex vivo* microscopy will be required to confirm whether the observed cluster spatially coincides with local retinal vasculopathy.

Despite these results, the extended 100° FOV introduces challenges, including off-axis aberrations, field curvature, and field-dependent signal variation. While quantitative evaluation is desirable, rigorous off-axis characterization using standard approaches [15] is non-trivial due to the limited working distance (~6.4 mm), which prevents sufficient target tilt while maintaining alignment and focus. Nevertheless, *in vivo* imaging demonstrates that key retinal features of interest, such as nerve fiber bundles, capillary networks, and microglial structures, remain resolvable across a wide spatial extent. This indicates that, despite peripheral degradation, the system maintains good image quality. Moreover, we performed an FFT-based modulation transfer function analysis to assess the field-dependent performance (Supplementary Fig. S2). The result shows that high spatial frequency responses decrease toward the periphery, while moderate spatial frequencies are well preserved to resolve fine features. Another limitation in this work is that OCT and SLO images are acquired sequentially, which may introduce minor spatial mismatches between modalities. In future work, we will explore a custom telescope to further suppress off-axis aberrations [15] and will also investigate a synchronized acquisition approach.

To sum up, we demonstrate a non-contact, multimodal vis-OCT and SLO system that achieves a 100° FOV for mouse retinal imaging. *En face* OCT and OCTA clearly visualized retinal structure and vasculature, while SLO resolves individual fluorescently labeled microglia. This platform enables studying retinal structural and vascular pathology, as well as microglia-related immune response *in vivo*. Further improvements, particularly in custom optics, should enhance image uniformity and quantitative reliability for longitudinal studies.

APPENDIX

See Supplementary material for supporting content.

CONFLICT OF INTEREST

Shuibin Ni: Siloam Vision; Tristan T. Hormel: IFOCUS IMAGING; Yukun Guo: Visonix/Optovue, Genentech; J. Peter Campbell: Siloam Vision, Genentech; Yifan Jian: Siloam Vision; Yali Jia: Visionix/Optovue, Roche/Genentech, IFOCUS IMAGING, Optos, Boeringer Ingelheim, Kugler. The potential conflicts of interest have been reviewed and managed by Oregon Health and Science University. Other authors declare no conflicts of interest related to this letter.

REFERENCES

- [1] T. Blank et al., "Early microglia activation precedes photoreceptor degeneration in a mouse model of CNGB1-linked retinitis pigmentosa," *Frontiers Immunology*, vol. 8, p. 1930, Jan. 2018, doi: [10.3389/fimmu.2017.01930](https://doi.org/10.3389/fimmu.2017.01930).
- [2] M. Grannonic et al., "Longitudinal analysis of retinal ganglion cell damage at individual axon bundle level in mice using visible-light optical coherence tomography fibergraphy," *Transl. Vis. Sci. Technol.*, vol. 12, no. 5, p. 10, May 2023, doi: [10.1167/tvst.12.5.10](https://doi.org/10.1167/tvst.12.5.10).
- [3] G. B. Liang et al., "112° field of view high-resolution swept-source OCT angiography for rat retinas," *Opt. Lett.*, vol. 49, no. 22, pp. 6449–6452, Nov. 2024, doi: [10.1364/ol.538165](https://doi.org/10.1364/ol.538165).
- [4] S. Pi et al., "Retinal capillary oximetry with visible light optical coherence tomography," *Proc. Nat. Acad. Sci. USA*, vol. 117, no. 21, pp. 11658–11666, May 2020, doi: [10.1073/pnas.1918546117](https://doi.org/10.1073/pnas.1918546117).
- [5] R. J. Zawadzki et al., "Adaptive-optics SLO imaging combined with widefield OCT and SLO enables precise 3D localization of fluorescent cells in the mouse retina," *Biomed. Opt. Exp.*, vol. 6, no. 6, pp. 2191–2210, Jun. 2015, doi: [10.1364/boe.6.002191](https://doi.org/10.1364/boe.6.002191).
- [6] T. Xu et al., "Integrated visible-light optical coherence tomography and fluorescence scanning laser ophthalmoscopy to image retinal ganglion cell axons," *Biomed. Opt. Exp.*, vol. 16, no. 7, pp. 2847–2856, Jul. 2025, doi: [10.1364/boe.561992](https://doi.org/10.1364/boe.561992).
- [7] E. B. Miller, P. Zhang, K. Ching, E. N. Pugh, and M. E. Burns, "In vivo imaging reveals transient microglia recruitment and functional recovery of photoreceptor signaling after injury," *Proc. Nat. Acad. Sci. USA*, vol. 116, no. 33, pp. 16603–16612, Aug. 2019, doi: [10.1073/pnas.1903336116](https://doi.org/10.1073/pnas.1903336116).
- [8] A. Joseph, D. Power, and J. Schallek, "Imaging the dynamics of individual processes of microglia in the living retina in vivo," *Biomed. Opt. Exp.*, vol. 12, no. 10, pp. 6157–6183, Oct. 2021, doi: [10.1364/boe.426157](https://doi.org/10.1364/boe.426157).
- [9] S. Song et al., "High-resolution multimodal visible light optical coherence tomography and scanning laser ophthalmoscopy for in vivo neuronal and vascular retinal imaging in mice," *Biomed. Opt. Exp.*, vol. 16, no. 6, p. 2365, Jun. 2025, doi: [10.1364/boe.560539](https://doi.org/10.1364/boe.560539).
- [10] N. Valikodath, E. I. Cole, M. F. Chiang, J. P. Campbell, and R. Chan, "Imaging in retinopathy of prematurity," *Asia-Pacific J. Ophthalmol.*, vol. 8, no. 2, pp. 178–186, Mar. 2019, doi: [10.22608/apo.201963](https://doi.org/10.22608/apo.201963).
- [11] Z. Tang et al., "An innovative multi-modal retinal imaging system for in vivo retinal detection in small animals," *Frontiers Ophthalmology*, vol. 3, Sep. 2023, Art. no. 1251328, doi: [10.3389/fopht.2023.1251328](https://doi.org/10.3389/fopht.2023.1251328).
- [12] S. Song et al., "100-degree non-contact visible-light OCT and SLO for in vivo mouse retinal imaging," *Proc. SPIE*, pp. 46–48, Mar. 2026, doi: [10.1117/12.3082391](https://doi.org/10.1117/12.3082391).
- [13] B. Hofer, B. Považay, B. Hermann, A. Unterhuber, G. Matz, and W. Drexler, "Dispersion encoded full range frequency domain optical coherence tomography," *Opt. Exp.*, vol. 17, no. 1, pp. 7–24, Jan. 2009, doi: [10.1364/oe.17.000007](https://doi.org/10.1364/oe.17.000007).
- [14] M. Pan, R. Meng, A. Gupta, and V. J. Srinivasan, "Transverse chromatic aberration (TCA) in visible light optical coherence tomography (OCT) of the human retina," *Proc. SPIE*, Mar. 2026, Art. no. PC138310V, doi: [10.1117/12.3085228](https://doi.org/10.1117/12.3085228).
- [15] S. Ni et al., "Non-mydratic handheld optical coherence tomography for panretinal imaging," *Biomed. Opt. Exp.*, vol. 16, no. 9, pp. 3665–3681, Sep. 2025, doi: [10.1364/boe.569243](https://doi.org/10.1364/boe.569243).